based on my resume, I have a Full Stack Intern role Interview today. It is Scheduled for 30 mins. So, Tell me the topics that I should prepare in advance for this role

Based on your resume, this is likely not going to be a generic "What is React?" interview. The interviewer will probably spend a good portion of the 30 minutes evaluating the technologies you specifically claim on your resume and how deeply you understand the projects you've built.

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Expected Interview Structure (30 min)

5 min: Introduction + resume walkthrough

15–20 min: Technical questions based on:

- React.js / Next.js
- JavaScript
- FastAPI & Backend
- REST APIs
- SQL & Databases
- Your MineAI project
- Basic DSA/problem solving

5 min: Your questions for interviewer

Priority 1: React + JavaScript (Highest Probability)

You mention React.js and Next.js prominently, so expect questions here.

React Fundamentals

Prepare:

- What is React?
- What is Virtual DOM?
- Difference between DOM and Virtual DOM
- What are Components?
- Functional vs Class Components
- What are Props?
- What is State?
- useState
- useEffect
- Component Lifecycle (especially useEffect)

Example questions:

What happens when state changes?
Why use useEffect?
Difference between props and state?

React Performance

Know:

- Re-rendering
- Keys in lists
- React.memo
- useMemo
- useCallback

Basic understanding is enough for intern level.

Next.js

You mention Next.js specifically.

Prepare:

- Why Next.js over React?
- SSR vs CSR
- Routing in Next.js
- API Routes
- Benefits of Next.js

Interview answer:

> Next.js provides file-based routing, server-side rendering, better SEO, API routes, and improved performance compared to plain React.

JavaScript Questions

Very common.

Know:

- var vs let vs const
- == vs ===
- Arrow functions
- Closures
- Promises
- async/await
- Event Loop (basic)

Example:

> Difference between synchronous and asynchronous code?
>
> What is a Promise?

Priority 2: FastAPI & Backend

Your MineAI project heavily uses FastAPI.

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FastAPI

Know:

- Why FastAPI?
- FastAPI vs Flask
- API endpoints
- Request/Response cycle
- Path parameters
- Query parameters
- Pydantic models

Example:

> Why did you choose FastAPI?

Good answer:

> FastAPI provides automatic validation through Pydantic, automatic OpenAPI documentation, type hints, and excellent performance.

REST APIs

Very likely.

Know:

Methods:

- GET
- POST
- PUT

- PATCH
- DELETE

Questions:

Difference between PUT and PATCH?

What is a REST API?

What status codes have you used?

Know:

- 200 OK
- 201 Created
- 400 Bad Request
- 404 Not Found
- 500 Internal Server Error

Priority 3: SQL & Databases

Your MineAI project uses SQLAlchemy and SQLite.

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SQL Basics

Prepare:

```
SQLSELECT
WHERE
GROUPBY
ORDERBY
JOIN
```

Questions:

Difference between INNER JOIN and LEFT JOIN?

Primary Key?

Foreign Key?

What is normalization?

SQLAlchemy

Basic understanding:

- ORM meaning
- Why ORM instead of raw SQL
- Models
- Sessions

Example:

> What is ORM?

Answer:

> ORM maps database tables to Python objects, reducing the need to write raw SQL queries.

Priority 4: Your MineAI Project (MOST IMPORTANT)

This project will probably generate the most questions.

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Be ready to explain:

Architecture

Draw mentally:

```
Frontend
   ↓
FastAPI
   ↓
LangChain
```

↓
FAISS / SQL
↓
OpenAI API

Questions They May Ask

What is RAG?

Answer:

> Retrieval-Augmented Generation retrieves relevant information from a knowledge base and provides it to the LLM before generating a response.

Why FAISS?

Answer:

> FAISS enables efficient vector similarity search over embeddings.

What are embeddings?

Answer:

> Embeddings are numerical vector representations of text that capture semantic meaning.

How does semantic search work?

Answer:

> User query is converted into an embedding and compared against stored embeddings using similarity search.

What is NL-to-SQL?

Answer:

> Converting natural language queries into SQL statements automatically.

Example:

> "Show sales from last month"

↓

```
SQLSELECT*FROM sales
WHEREmonth='last_month';
```

Priority 5: Git & GitHub

You list Git/GitHub in skills.

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Know:

```
Bashgit clone
git add
git commit
git push
git pull
git branch
git merge
```

Questions:

> Difference between pull and fetch?
>
> What is a branch?

Priority 6: Basic DSA

Many companies ask 1-2 quick questions.

Prepare:

Arrays

- Reverse array
- Find maximum

- Remove duplicates

Strings

- Reverse string
- Palindrome

Hash Maps

- Frequency counting

Complexity

Know:

- O(1)
- O(log n)
- O(n)
- O(n^2)

Question:

> Difference between Array and HashMap?

Priority 7: AI/ML Questions

Since you're an AI/ML student and mention Scikit-learn.

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Prepare:

Machine Learning Basics

- Supervised Learning
- Unsupervised Learning
- Classification
- Regression

Algorithms

You mention:

- Random Forest
- Logistic Regression

Know:

Why Random Forest?
Why Logistic Regression?
Overfitting?
Train/Test Split?

HR Questions (Very Likely)

Prepare concise answers:

Tell me about yourself

90-second version:

- AI/ML student at DJSCE
- Full-stack development experience
- Built MineAI using FastAPI + LangChain + React/Next.js
- Interested in backend systems and scalable products
- Looking to learn and contribute as a Full Stack Intern

Why do you want this internship?

Focus on:

- Real-world experience
- Working on production systems
- Learning from experienced engineers

- Applying full-stack skills

Tell me about a challenge you faced

Use MineAI:

- Integrating vector search + SQL
- Handling API responses
- Debugging backend issues
- Optimizing retrieval quality

If You Only Have 1 Hour Before the Interview

Prioritize in this order:

1. Explain every line of MineAI project
2. React (state, props, useEffect)
3. FastAPI + REST APIs
4. SQL joins
5. JavaScript async/await
6. Git basics
7. Complexity O(n), O(log n)

For an intern role, being able to confidently explain your projects is usually more important than knowing advanced system design. If they ask about anything on your resume, you should be able to explain **why you used it, how it works, and one challenge you faced with it**.

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help me prepare this all topics fully focused on techincal round interview prep in detail.

Given your resume and a **30-minute Full Stack Intern interview**, I would focus on what interviewers actually ask rather than trying to learn every topic in the ecosystem.

Your preparation should be centered around:

1. JavaScript
2. React
3. Next.js
4. FastAPI
5. REST APIs
6. SQL
7. Git
8. Your MineAI project (most important)
9. Basic DSA

Everything below is optimized for interview answers.

SECTION 1: JAVASCRIPT

Q1. Difference between var, let and const

var

JavaScriptvarx=10;

- Function scoped
- Can be redeclared
- Can be updated

```
JavaScriptvarx=10;
varx=20;
```

Valid.

---

let

```
JavaScriptletx=10;
```

- Block scoped
- Cannot be redeclared
- Can be updated

```
JavaScriptletx=10;
x=20;
```

Valid.

---

const

```
JavaScriptconstx=10;
```

- Block scoped
- Cannot be redeclared
- Cannot be reassigned

```
JavaScriptconstx=10;
x=20;
```

Error.

Interview Answer

> var is function-scoped and allows redeclaration. let and const are block-scoped. const prevents reassignment while let allows updates.

Q2. == vs ===

JavaScript5=="5"

Returns:

JavaScripttrue

Type conversion happens.

JavaScript5==="5"

Returns:

JavaScriptfalse

Checks:

- Value
- Type

Interview Answer

> == performs type coercion while === checks both value and datatype.

Q3. What is a Closure?

```
JavaScriptfunctionouter() {
letcount=0;

returnfunctioninner() {
count++;
returncount;
  };
}
```

Closure means:

> A function remembers variables from its outer scope even after the outer function has finished executing.

Q4. Promise

Problem:

```
JavaScriptfetchData();
```

takes time.

Promise has:

```
JavaScriptPending
Resolved
Rejected
```

Example:

```
JavaScriptfetch(url)
.then(res => res.json())
.catch(err => console.log(err));
```

Q5. Async Await

Instead of:

```
JavaScript.then()
```

use:

```
JavaScriptasyncfunctiongetData() {
constres=awaitfetch(url);
constdata=awaitres.json();
}
```

Interview Answer

> async/await is syntactic sugar over promises and makes asynchronous code easier to read.

SECTION 2: REACT

What is React?

Interview answer:

> React is a JavaScript library used for building reusable user interfaces through components.

What is Virtual DOM?

Normal DOM updates are expensive.

React:

1. Creates Virtual DOM
2. Compares old and new Virtual DOM
3. Updates only changed elements

Called:

Diffing
Reconciliation

What is JSX?

```
JavaScriptconstelement=<h1>Hello</h1>;
```

JSX allows writing HTML inside JavaScript.

Props

```
JavaScriptfunctionUser(props) {
return<h1>{props.name}</h1>;
}
```

Props:

- Passed from parent
- Read only

State

JavaScriptconst [count, setCount] =useState(0);

State:

- Internal component data
- Causes re-render

Props vs State

Props	State
Parent passes	Component manages
Read only	Mutable
External	Internal

useState

JavaScriptconst [count, setCount] =useState(0);

Updates UI automatically.

---

useEffect

JavaScriptuseEffect(() => {
fetchData();
}, []);

Runs:

- After render
- For API calls

- Side effects

Why useEffect?

Examples:

- Fetching data
- Timers
- Event listeners

Component Lifecycle

Mount

Component appears.

JavaScriptuseEffect(() => {
}, []);

Update

State changes.

Unmount

Cleanup.

JavaScriptreturn () => {
clearInterval(timer);
};

React Keys

Bad:

```
JavaScriptitems.map(item => <li>{item}</li>)
```

Good:

```
JavaScriptitems.map(item => (
<likey={item.id}>{item.name}</li>
))
```

Keys help React identify changed elements.

SECTION 3: NEXT.JS

You mention Next.js, so expect questions.

Why Next.js?

React only:

Frontend Library

Next.js adds:

- Routing
- SSR
- SEO
- API routes
- Better performance

SSR vs CSR

CSR

Browser renders page.

Examples:

React SPA

Pros:

- Fast navigation

Cons:

- SEO issues

SSR

Server renders HTML first.

Pros:

- Better SEO
- Faster first load

Interview Answer

> SSR renders pages on the server before sending them to the browser, while CSR renders them in the browser using JavaScript.

SECTION 4: FASTAPI

What is FastAPI?

Interview answer:

FastAPI is a modern Python backend framework for building APIs using type hints and automatic validation.

Why FastAPI?

Advantages:

- Fast
- Async support
- Auto documentation
- Pydantic validation

Endpoint Example

PythonRun@app.get("/users")
defget_users():
returnusers

Path Parameters

PythonRun@app.get("/users/{id}")

Example:

/users/1

Query Parameters

PythonRun@app.get("/users")

Example:

/ users?page=2

Pydantic

PythonRunclassUser(BaseModel):
name: str
age: int

Validation happens automatically.

SECTION 5: REST API

What is REST?

REST is a standard architecture for communication between frontend and backend.

GET

Retrieve data.

httpGET /users

POST

Create data.

httpPOST /users

PUT

Replace entire object.

PATCH

Update only specific fields.

DELETE

Remove resource.

PUT vs PATCH

PUT:

```
JSON{
  "name":"Pranav",
  "age":20
}
```

Full update.

PATCH:

```
JSON{
  "age":21
}
```

Partial update.

HTTP Status Codes

200 OK
201 Created
400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
500 Internal Server Error

Must know these.

SECTION 6: SQL

Primary Key

Uniquely identifies row.

SQLid INTPRIMARYKEY

Foreign Key

Links tables.

SQLuser_id

references Users table.

INNER JOIN

SQLSELECT*
FROM Orders
INNERJOIN Users
ON Orders.user_id = Users.id;

Returns matching rows.

LEFT JOIN

Returns:

- All rows from left table
- Matching rows from right

Difference

INNER JOIN:

Only matches.

LEFT JOIN:

Keeps all left rows.

GROUP BY

```sql
SQLSELECT department,
COUNT(*)
FROM employees
GROUPBY department;
```

ORDER BY

```sql
SQLSELECT*
FROM employees
ORDERBY salary DESC;
```

SECTION 7: GIT

---

git clone

Downloads repository.

---

git add

Stage changes.

```bash
Bashgit add .
```

---

git commit

Create snapshot.

Bashgit commit -m"added feature"

---

git push

Upload changes.

---

git pull

Download latest changes.

Branch

Bashgit checkout -b feature

Used for isolated development.

SECTION 8: MINEAI PROJECT (MOST IMPORTANT)

Your interviewer will likely spend the most time here.

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You must explain:

What problem does MineAI solve?

Answer:

> MineAI allows users to query structured and unstructured data using natural language instead of manually searching documents or writing SQL queries.

Complete Flow

User Query
↓
FastAPI
↓
LangChain
↓
Embedding Model
↓
FAISS Search
↓
Relevant Documents
↓
OpenAI
↓
Response

What is RAG?

Interview answer:

> RAG stands for Retrieval Augmented Generation. Before generating a response, the system retrieves relevant information from a knowledge base and provides it as context to the LLM.

What are Embeddings?

Answer:

> Embeddings are numerical vector representations of text that capture semantic meaning.

Example:

```
Car
Vehicle
Automobile
```

have similar vectors.

What is FAISS?

Answer:

> FAISS is a vector database library used for efficient similarity search among embeddings.

What is Semantic Search?

Keyword search:

```
car
```

matches only "car".

Semantic search:

```
automobile
```

can also find "car".

What is LangChain?

Answer:

> LangChain provides tools and abstractions for connecting LLMs with external data sources, retrieval systems and workflows.

What is NL-to-SQL?

Question:

> Show top customers from Mumbai.

Generated SQL:

```
SQLSELECT*
FROM customers
WHERE city='Mumbai';
```

Natural language converted to SQL automatically.

SECTION 9: DSA

Know complexity.

O(1)

```
JavaScriptarr[0]
```

O(n)

```
JavaScriptfor(leti=0;i<n;i++)
```

$O(n^2)$

Nested loops.

Reverse String

```
JavaScriptstr.split('').reverse().join('')
```

Palindrome

```
JavaScriptfunctionisPalindrome(str){
returnstr===str.split('').reverse().join('');
}
```

Questions You Should Ask Them

At the end:

1. What technologies does the team primarily use?
2. What does success look like for an intern in the first 3 months?
3. What kind of projects would I be contributing to?

Final 30-Minute Revision Checklist

Memorize answers for:

✅ What is React?

✅ Props vs State

✅ useState

✅ useEffect

✅ Virtual DOM

✅ SSR vs CSR

✅ FastAPI advantages

✅ REST API methods

✅ PUT vs PATCH

✅ Primary Key vs Foreign Key

✅ INNER JOIN vs LEFT JOIN

✅ What is RAG?

✅ What is FAISS?

✅ What are Embeddings?

✅ Explain MineAI end-to-end

✅ async/await

✅ var vs let vs const

✅ Time Complexity O(1), O(n), O(n^2)

If you can confidently answer these without hesitation, you'll be well prepared for the majority of Full Stack Intern technical interviews based on your current resume.

Pranav Khandelwal